Security Analysis Report

Objective

The objective of this phase was to evaluate the security and privacy of the proposed privacy-preserving iris authentication system. Different attack scenarios were simulated to analyse how well the encrypted biometric templates are protected against unauthorized access and misuse.

1) Template Theft Attack

In this attack, it was assumed that an attacker gains access to the encrypted biometric database. The encrypted templates generated using CKKS homomorphic encryption were analyzed. The stolen data consisted only of encrypted ciphertext and no readable biometric information was visible. The ciphertext appeared as random byte data and did not reveal any iris features.

Observation

1) Encrypted templates were successfully protected.
2) No raw biometric information was exposed.
3) Privacy leakage was not observed.

2) Insider Attack

In this scenario, a database administrator was assumed to have access to the encrypted database. The encrypted templates could be viewed, but the original iris embeddings could not be accessed directly. Since the biometric templates were stored in encrypted form, the administrator could not obtain the actual iris information.

Observation

1) Encrypted templates were accessible.
2) Raw biometric data was not accessible.
3) User privacy remained protected.

3) <u>Replay Attack</u>

A replay attack was simulated by reusing an already stored encrypted template during authentication. The similarity score obtained was approximately equal to 1 because the same template was compared with itself.

<u>Observation</u>

1) Replay attacks can produce high similarity scores.
2) Template encryption alone does not completely prevent replay attacks.

4) <u>Reconstruction Attack</u>

An attempt was made to analyze whether the original iris image could be reconstructed from the encrypted embeddings. Since only encrypted feature vectors were available and no direct biometric information was exposed, reconstruction of the original iris image was not possible.

<u>Observation</u>

1) Iris reconstruction was unsuccessful.
2) Encrypted embeddings did not reveal image information.
3) Biometric privacy was preserved.

<u>Security Analysis</u>

	Attack	Result
0	Template Theft	Encrypted Templates Only
1	Insider Attack	No Raw Biometric Exposure
2	Replay Attack	Replay Detected Through Matching
3	Reconstruction Attack	Reconstruction Failed

Security Evaluation Summary

The security analysis demonstrated that the proposed system provides strong protection for biometric templates. The use of CKKS homomorphic encryption ensured that biometric data remained encrypted during storage and matching operations. Template theft and insider attacks did not expose raw biometric information, while reconstruction of iris images was not possible from encrypted data.

Conclusion

-> Template theft attack was successfully resisted.
-> Insider access did not reveal biometric information.
-> Replay attacks may require additional protection mechanisms.
-> Reconstruction of iris images from encrypted embeddings failed.
-> The proposed privacy-preserving iris authentication system provides strong security and privacy protection for biometric data.